Generation Probabilities are Not Enough: Improving Error Highlighting in AI Code Suggestions

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Programmers increasingly use AI generated code (e.g., from CoPilot) in IDEs!

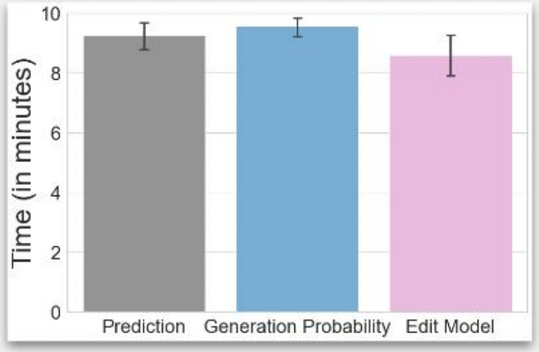
However, generated code may be erroneous! ***We need to help programmers double-check AI code and identify when and where it may be incorrect.***

We explore two options to highlight uncertain, potentially erroneous regions:

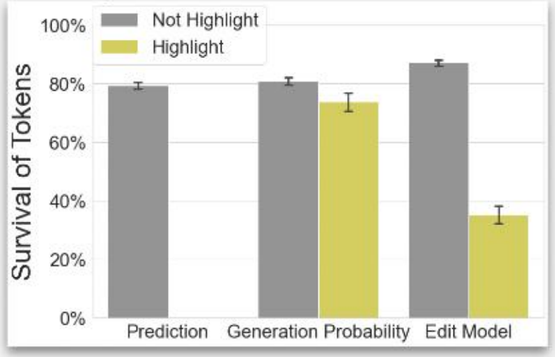
- *Use* **generation probabilities** *or "confidence scores" to show where the model is uncertain*
- Learn an **edit model** to show where programmers are most likely to make edits

Mixed-methods, within-subjects study with 30 programmers showed that ***edit model leads to***:

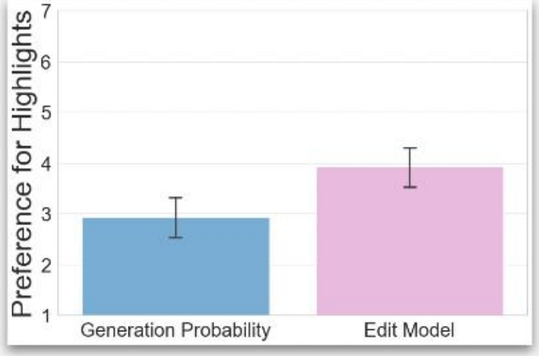
- Significantly ***faster*** task completion time



- Significantly ***more localized edits***



- Stronger ***preference***



compared to the generation probability and no-highlights